The architecture of modern education has undergone a fundamental transformation, shifting from an input-based, instructor-centric model to an output-based, learner-centric paradigm. This evolution is centered on the concept of learning outcomes, which serve as the definitive statements of what a learner is expected to know, understand, and demonstrate upon the successful completion of a learning process.[1] Unlike traditional instructional objectives, which frequently describe the teacher's intentions or the scope of the curriculum, learning outcomes are focused strictly on the demonstrable achievement of the student.[2] This shift represents a commitment to transparency, accountability, and the optimization of cognitive and affective development across the European Higher Education Area (EHEA) and other international jurisdictions.[4]

Theoretical Foundations and Taxonomies of Learning

The effective articulation of learning outcomes requires a rigorous theoretical framework that categorizes the complexity of human cognition, skill acquisition, and attitudinal change. These taxonomies provide the necessary vocabulary and structural hierarchy to ensure that educational goals are both developmentally appropriate and intellectually rigorous.[7]

Bloom's Revised Taxonomy and the Cognitive Domain

The most pervasive framework in the design of learning outcomes is Bloom's Taxonomy, originally established in 1956 and substantively revised in 2001 by Anderson and Krathwohl. This revised model organizes cognitive processes into a six-level hierarchy: remember, understand, apply, analyze, evaluate, and create.[7] The hierarchical nature of the taxonomy suggests that higher-order thinking skills, such as creation and evaluation, are progressively built upon the foundation of lower-order skills like recall and comprehension.[7] For instance, a student cannot effectively analyze a complex political system without first remembering the historical facts of its development and understanding its core theoretical principles.[8]

A significant departure in the 2001 revision was the elevation of "creating" to the highest level of the pyramid, replacing "evaluation," which was considered the pinnacle in the original 1956 version.[9] This change reflects a modern pedagogical emphasis on synthesis and the production of new knowledge as the ultimate goal of higher education.[9]

Cognitive Level	Rationale and Definition	Typical Observable Behaviors
Remember	Recall facts, basic concepts, and data from long-term memory.[7]	Defining, listing, memorizing, repeating, stating.[7]

Understand	Make meaning of information; explain ideas or concepts.[7]	Classifying, describing, discussing, explaining, identifying, locating.[7]
Apply	Use information and concepts in new or familiar situations.[7]	Executing, implementing, solving, demonstrating, interpreting, operating.[7]
Analyze	Break material into subcomponents and determine relationships.[7]	Differentiating, organizing, relating, comparing, contrasting, examining.[7]
Evaluate	Make judgments based on criteria and standards; justify a decision.[7]	Appraising, arguing, defending, judging, critiquing, weighing.[7]
Create	Organize elements into a new pattern or structure; produce original work.[7]	Designing, assembling, constructing, formulating, authoring, investigating.[7]

Beyond the cognitive domain, Bloom's framework extends into the affective and psychomotor domains. The affective domain addresses the development of interests, attitudes, and values—elements that are often present in an instructor's "secret" or "embedded" intent but are rarely articulated in a formal syllabus.[12] The psychomotor domain focuses on physical coordination and motor skills, which are essential in clinical, technical, and vocational disciplines.[9]

Fink's Taxonomy of Significant Learning

While Bloom's Taxonomy provides a robust structure for cognitive development, L. Dee Fink's Taxonomy of Significant Learning (2003) offers a more holistic and relational approach. Fink identifies six interrelated categories that describe how learning can create lasting change in multiple dimensions of a learner's life.[7] These categories include foundational knowledge, application, integration, human dimension, caring, and learning how to learn.[9]

The "human dimension" and "caring" categories are particularly distinctive, as they highlight the social and emotional impact of education. These domains encourage students to learn about themselves and others while developing new feelings or values regarding the subject matter.[7] Furthermore, the "learning how to learn" category focuses on metacognition, which aims to transform students into self-directed, lifelong learners capable of navigating information independently.[9]

The Intersection of Cognitive and Holistic Taxonomies

Many institutions now advocate for the complementary use of both Bloom's and Fink's frameworks. This approach allows educators to leverage Bloom for structural cognitive scaffolding while using Fink to ensure broader educational impact and deeper student engagement.[7] The intersection of these taxonomies allows for the selection of specific action verbs that align cognitive complexity with holistic significance.[10]

Bloom's Level	Fink's Domain: Human Dimension	Fink's Domain: Caring	Fink's Domain: Integration
Create	Advocate, Propose, Build, Role play.[10]	Generate, Improve, Synthesize, Cultivate.[10]	Construct, Predict, Integrate, Host.[10]
Evaluate	Critique, Clarify, Support, Exemplify.[10]	Assess, Judge, Reconcile, Recommend.[10]	Conclude, Associate, Test, Grade.[10]
Analyze	Characterize, Distinguish, Mediate, Discover.[10]	Explore, Correlate, Empathize, Assemble.[10]	Question, Relate, Tabulate, Dismantle.[10]
Apply	Demonstrate, Promote, Associate, Cooperate.[10]	Transfer, Express, Share, Initiate.[10]	Use, Personalize, Combine, Recommend.[10]
Understand	Represent, Compare, Translate, Discuss.[10]	Select, Interpret, Respond, Value.[10]	Extend, Convert, Plot, Visualize.[10]
Remember	Specify, Recognize, Label, Accept.[10]	Quote, Recall, Copy, Adhere.[10]	Indicate, Recite, Blend, Trace.[10]

Structural Mechanics of Articulating Outcomes

The technical precision with which a learning outcome is drafted determines its ultimate utility as a tool for assessment and student guidance. Poorly articulated outcomes often lead to vague assessments and a lack of clarity in the learning pathway, whereas precise outcomes provide a definitive blueprint for evaluation.[13]

Distinguishing Outcomes from Objectives and Goals

In professional course design, it is essential to distinguish between learning goals, learning

objectives, and learning outcomes, although these terms are frequently used interchangeably in informal discourse. Learning goals are broad, general statements of teaching intention, often describing what the instructor or the program aims to accomplish (e.g., "The curriculum will introduce students to major research methods").[2]

Learning objectives are more specific than goals and typically describe the instructor's planned coverage of content or the steps leading toward an outcome. They are instructor-centered and reflect the *means* rather than the *ends*.[3] In contrast, learning outcomes are student-centered, observable, and measurable statements that describe what the student will be able to *do* as a result of the learning experience.[3]

The SMARTIE and ABCD Frameworks

Best practices for drafting outcomes suggest that they should be specific, measurable, achievable, relevant, and time-bound (SMART). Recent refinements have added "Inclusive" and "Evident" to create the SMARTIE framework.[17] Inclusivity ensures that the outcome is designed using universal design principles to accommodate diverse learners, while "Evident" ensures that the learning is visible and transparent to the students themselves, promoting metacognition.[13]

The ABCD model provides a further structural formula for writing specific outcomes:

- **Audience**: Identifies who will achieve the outcome (e.g., "The student" or "The participant").[11]
- **Behavior**: Employs a measurable action verb to describe the performance.[11]
- **Condition**: Specifies the circumstances, context, or tools under which the behavior is performed (e.g., "Using statistical software" or "In a simulated clinical environment").[11]
- **Degree**: Defines the standard or criteria for success (e.g., "With 95% accuracy" or "According to professional industry standards").[11]

The Critical Role of Action Verbs

The selection of the verb is the most vital element of a learning outcome. Educators must avoid vague or unmeasurable terms such as *understand*, *know*, *be aware of*, *appreciate*, *learn*, or *become familiar with*.[19] These verbs describe internal mental states that cannot be directly observed or verified.[21] Instead, measurable verbs such as *analyze*, *critique*, *design*, *state*, *summarize*, or *calculate* should be utilized.[13] If the term "understand" must be used, it should be framed as "demonstrate an understanding of X by doing Y," where Y is a specific, measurable action.[12]

Constructive Alignment and Curricular Integration

A learning outcome does not function in isolation; it is a critical component of a "Golden Triangle" that includes teaching and learning activities and assessment tasks.[8] This concept,

pioneered as "Constructive Alignment" by John Biggs, posits that for learning to be effective, all components of the teaching system must be deliberately aligned to support the student in achieving the intended outcomes.[16]

The Alignment Triangle in Course Design

Alignment is achieved when the learning outcomes define the intended goal, the instructional strategies foster the necessary skills and knowledge, and the assessments provide evidence of whether those goals were met.[22] Misalignment occurs, for example, when an outcome requires "critical analysis," but the assessment only measures "factual recall" through a multiple-choice quiz.[24] Such discrepancies lead to student frustration and a failure to capture true intellectual competence.[24]

Outcome Level	Appropriate Instructional Strategies	Appropriate Assessment Methods
Recall / Recognize	Lectures, readings, flashcards.[24]	Multiple-choice, matching, labeling.[24]
Interpret / Explain	Discussions, concept mapping, summarizing.[24]	Papers, class discussions, paraphrasing.[24]
Apply / Execute	Practice sets, labs, simulations.[24]	Problem sets, performances, prototyping.[24]
Analyze / Differentiate	Case studies, debates, critiques.[24]	Case studies, papers, projects.[24]
Evaluate / Critique	Peer review, product reviews, debates.[24]	Journals, critiques, product reviews.[24]
Create / Design	Research projects, composition, planning.[24]	Portfolios, research projects, compositions.[24]

Curriculum Mapping and Programmatic Approaches

At the institutional and program levels, individual course outcomes must map onto broader program-level learning outcomes (PLOs). Curriculum mapping is a systematic process that creates a visual roadmap showing where each PLO is taught, reinforced, and assessed across a degree program.[25] These maps typically utilize the "I-R-A" model:

- **Introduced (I)**: The stage where skills or knowledge are first presented to the student.[25]

- **Reinforced (R)**: The stage where the learning is practiced and built upon in subsequent units.[25]
- **Assessed (A)**: The stage where the student is required to demonstrate mastery, often for the purpose of award classification.[25]

Strategic curriculum mapping prevents "curriculum fragmentation" and ensures that students encounter a coherent pathway that builds complexity progressively.[26] It also identifies "milestones" or "stage-gate assessments" where progression is validated before a student is permitted to advance.[26] In the contemporary era of generative artificial intelligence, a "two-lane" approach to assessment has emerged: Lane 1 involves secured, in-person assessments to validate learning at critical program milestones, while Lane 2 involves open, unit-level assessments where students are taught to use AI tools responsibly in their learning process.[26]

Evaluation of Learning Outcomes: Evidence and Standards

The evaluation of learning outcomes requires the collection of evidence that is both valid and reliable. Validity refers to whether an assessment actually measures the specific knowledge or skill it was intended to measure, while reliability refers to the consistency of those measurements across different assessors and over time.[17]

Direct vs. Indirect Evidence of Student Learning

High-quality evaluation strategies employ a combination of direct and indirect evidence to gain a comprehensive understanding of student achievement.[30] Direct evidence is tangible and observable, involving the direct examination of student work.[30] Indirect evidence reflects the perceptions or attitudes of the students or other stakeholders.[30]

Evidence Category	Sources and Examples	Rationale and Utility
Direct Evidence	Portfolios, capstone projects, licensure exams, rubric-scored performances.[30]	Provides compelling, actionable data on actual student capabilities.[30]
Indirect Evidence	Student surveys, focus groups, alumni interviews, course evaluations.[30]	Explains the "why" behind the results and provides context on student satisfaction.[31]

It is a common misconception that course grades are a direct measure of specific learning outcomes. In technical terms, grades are considered "indirect" measurements because they typically represent an aggregate of multiple outcomes and may be influenced by factors unrelated to learning, such as attendance or extra credit.[30]

Rubrics and Standardized Assessment Frameworks

To ensure consistency and fairness in evaluation, the use of rubrics is considered a global best practice. Rubrics break down complex outcomes into specific criteria and performance levels, allowing for more objective and efficient grading.[34] The AAC&U VALUE rubrics (Valid Assessment of Learning in Undergraduate Education) are a widely recognized national and international standard for assessing complex outcomes like critical thinking, ethical reasoning, and global learning.[36]

These rubrics are intended for institutional discussion and the evaluation of authentic student work, such as research papers and projects, rather than simple standardized testing.[37] They help clarify faculty expectations and provide credible evidence that can be used for continuous program improvement and accountability.[37]

Institutional Management and Regulatory Frameworks

The implementation of learning outcomes is increasingly governed by national and international regulatory standards. In the Irish context, the Quality and Qualifications Ireland (QQI) provides the statutory basis for outcomes-based program development.[1] The National Framework of Qualifications (NFQ) defines 10 levels of learning in terms of three strands: knowledge, skills, and competence.[40]

The European Credit Transfer and Accumulation System (ECTS)

The ECTS Users' Guide serves as the primary framework for credit transfer and program design within the EHEA.[4] ECTS credits are based on the student workload required to achieve specific outcomes, with a full-time academic year generally equating to 60 credits.[4] Crucially, credits are only awarded after the "appropriate assessment" of the achieved learning outcomes.[5] The guide recommends that a module should have between 6 and 8 outcomes to ensure they remain "verifiable and achievable" within the allocated workload.[4]

Grading Conventions and Quality Assurance

Under QQI standards, assessment must be professionally managed and integrated into broader quality assurance procedures.[29] This includes the appointment of at least one independent expert external examiner for each award program to ensure the validity and reliability of the assessment process.[29]

Grading System	Pass/Fail Thresholds	Calculations and Conversions
Percentage	Passing marks are $40 \leq p \leq$.[29]	Marks below 35 ($p <$) are outright failures with a percentage point value (ppv) of zero.[29]
Alphabetic	Grades range from A (4.0) to D (1.5) for passing.[29]	A minimum Grade Point Average (GPA) of 2.0 is generally required to pass a stage.[29]

Case Studies and Sectoral Implementation

Sectoral implementations reveal that the success of learning outcomes often depends on active, context-specific pedagogical strategies.

Multi-Disciplinary and Integrative Strategies

Practical applications across diverse fields demonstrate how learning outcomes can bridge the gap between theoretical knowledge and professional practice:

- **Engineering Design**: The University of Bristol's Engineering Design program utilizes authentic industry engagement and open-ended projects to foster skills in creativity and entrepreneurship. This approach ensures students understand how technical issues interlink with social and environmental constraints.
- **Interdisciplinary STEMM Integration**: Research into the integration of arts and humanities into STEMM curricula shows significant gains in critical thinking and higher-order reasoning. For instance, undergraduate students have used haiku-style writing to articulate complex neurobiological concepts, such as addiction, facilitating deeper synthesis and meaning.
- **Business Education Mapping**: Case studies from the Irish National Forum demonstrate the value of program-level mapping, even in large-scale international programs such as Singapore-based Bachelor of Business Studies.[42] These maps reveal how expectations for assessment align with reality across the undergraduate student lifecycle.[42]

Digital Literacy and AI in Global Universities

Elite institutions are increasingly framing digital proficiency as a core learning outcome:

- **Harvard University**: Prioritizes digital literacy through a "triple logical framework" (value,

technological, and practical). Courses like CS50 provide real-time, AI-powered coding feedback, allowing students to achieve specific outcomes in data literacy and problem-solving through continuous, interactive feedback mechanisms.
- **Tsinghua University**: Employs a "Smart Campus" system (Rain Classroom) that uses AI-based analytics to identify and remove learning barriers. This system allows for personalized feedback loops that enhance the achievement of ethical digital citizenship outcomes.

Clinical and Vocational Excellence

- **Nursing Education**: Descriptive qualitative research in undergraduate nursing shows that case study analysis reduces the "theory-practice gap" during clinical placements.[44] Students engaged in this "learning by doing" reported significantly higher self-confidence and improved interpersonal relations compared to traditional lecture-based learning.
- **Problem-Based Learning (PBL)**: Implementation of PBL in engineering (Spain) and nursing (South Africa) has been shown to stimulate deeper learning approaches when combined with formative assessment by supervisors. However, some nursing professionals noted a perceived lack of basic science knowledge when PBL was used in isolation, highlighting the need for a balanced curriculum.

Comparative Analysis of Learning Outcome Quality

The literature identifies clear benchmarks for distinguishing between high-quality and substandard learning outcomes. Research suggests that outcomes must avoid instructor-centered delivery descriptions and unobservable mental states.[14]

Poor versus Excellent Learning Outcome Examples

The following table provides concrete examples from educational literature comparing weak or flawed outcomes with their improved counterparts.

Status	Outcome Statement	Assessment / Critique
Poor	"The student will successfully perform heart surgery on a patient."	**Unrealistic**: Represents a level of learning that is unattainable within the training conditions.
Excellent	"The student will be able to assess the vitals of a patient."	**Realistic**: Matches the instructional activities and is achievable for the level.

Poor	"Students will be able to name, explain, and apply..."[14]	**Confusing**: Uses multiple independent behaviors; unclear which one is assessed or how they are weighted.[14]
Excellent	"Students will summarize [X]..." or "Students will solve..."[14]	**Precise**: Uses a single measurable behavior per outcome for valid assessment.[14]
Poor	"Adjust the page layout settings in a spreadsheet."[45]	**Too Narrow**: Focuses on a minor task rather than a meaningful skill.[45]
Excellent	"Format spreadsheets to look professional in both hard and soft copy."[45]	**Appropriate Scope**: Focuses on a comprehensive, professional skill.[45]
Poor	"Students will understand marketing principles."[46]	**Vague**: "Understand" is not observable. Marketing messages to students become unclear.[46]
Excellent	"Students will develop and present a marketing plan for peer feedback."[46]	**Measurable**: Clearly aligns the outcome with a specific, observable assessment task.[46]
Poor	"Be aware of the main sources of research literature in their field."[17]	**Novice/Weak**: "Be aware" is a passive state that is difficult to test reliably.[17]
Excellent	"Cite the main sources of research literature in their field using APA format."[17]	**Actionable**: Defines the specific performance required to show awareness.[17]

Common Pitfalls and Strategic Solutions

- **Vague Verbs**: Ambiguous terms like "know," "be familiar with," or "grasp" lead to unreliable assessments because they do not specify what a learner must do.[46]

- **Instructor-Centeredness**: Describing what the *program* or *teacher* does (e.g., "The course will provide an introduction to...") fails to measure student achievement.[21]
- **Assessment Misalignment**: A frequent error is listing a practical outcome (e.g., "develop a plan") but using a non-practical assessment (e.g., a multiple-choice quiz).[46]
- **Cognitive Load**: Learning science indicates working memory handles only 4-7 "chunks" of information at once.[18] Over-specifying too many outcomes can overwhelm this "bottleneck," whereas grouping related outcomes into "schemas" supports retention.[18]

Synthesis and Future Outlook

The rigorous design and evaluation of learning outcomes are the bedrock of a transparent, effective, and accountable educational system. By shifting the focus from "time served" to "demonstrable achievement," institutions can better prepare students for the complexities of the modern global workforce.[2]

As the educational landscape continues to be reshaped by digital transformation and artificial intelligence, the role of learning outcomes becomes even more paramount. They provide the stable benchmarks against which the efficacy of new pedagogical tools can be evaluated. The future of the field lies in the integration of holistic domains—such as interpersonal attitudes and metacognition—with rigorous cognitive standards, ensuring that education remains not only a process of information transfer but a transformative and significant life experience.[7] Through systematic curriculum mapping and a commitment to constructive alignment, educators can ensure that the "intended" outcomes of an institution become the "actual" capabilities of its graduates.[13]

Works cited

1. TUS Guide to Writing and Using Learning Outcomes 2025 - 2030, accessed March 30, 2026, https://tus.ie/app/uploads/AcademicServices/Quality/Handbook/VOL6/3_TUS-Guide-to-Writing-and-Using-Learning-Outcomes-AC-17.10.2025.pdf
2. International Experience in the Implementation of Learning Outcomes in Higher Education - Documents & Reports, accessed March 30, 2026, https://documents1.worldbank.org/curated/en/936911559551873438/pdf/International-Experience-in-the-Implementation-of-Learning-Outcomes-in-Higher-Education-Case-Studies-of-Good-Practice.pdf
3. Course Objectives & Learning Outcomes | Course Design | Teaching Guides | Teaching Commons | DePaul University, Chicago, accessed March 30, 2026, https://resources.depaul.edu/teaching-commons/teaching-guides/course-design/Pages/course-objectives-learning-outcomes.aspx
4. ECTS Users' Guide, accessed March 30, 2026, https://www.internationalisering.vluhr.be/files/2-ECTS-Users-Guide_Klara-Engels-Perenyi.pdf
5. ECTS USERS' GUIDE, accessed March 30, 2026,

https://www.educacionmedica.net/pdf/documentos/espacio/ECTS_guide.pdf
6. ECTS User Guide Revision - University of Twente, accessed March 30, 2026, https://www.utwente.nl/en/education/student-services/news-events/news/2025/4/260815/ects-user-guide-revision
7. Creating Learning Outcomes | Center for Integrated Professional Development | Illinois State, accessed March 30, 2026, https://prodev.illinoisstate.edu/instructional-resources/design/learning-outcomes/
8. Connect Course Learning Outcomes to Assignments and Exams | Robert Gillespie Academic Skills Centre - University of Toronto Mississauga, accessed March 30, 2026, https://www.utm.utoronto.ca/rgasc/faculty-instructors/teaching-and-learning-resource-hub/connect-course-learning-outcomes-assignments
9. Frameworks & Taxonomies of Learning | Course Design | Teaching Guides, accessed March 30, 2026, https://resources.depaul.edu/teaching-commons/teaching-guides/course-design/Pages/teaching-learning-frameworks.aspx
10. Learning Outcomes & Objectives Framework – A Crossroad ..., accessed March 30, 2026, https://effectiveness.syr.edu/wp-content/uploads/2024/02/Outcome-Framework.pdf
11. Creating Learning Outcomes - Stanford Teaching Commons, accessed March 30, 2026, https://teachingcommons.stanford.edu/teaching-guides/foundations-course-design/course-planning/creating-learning-outcomes
12. Outcomes and Objectives - Teaching@Purdue, accessed March 30, 2026, https://www.purdue.edu/innovativelearning/teaching/module/outcomes-and-objectives/
13. Learning Objectives & Outcomes – Teaching Resources, accessed March 30, 2026, https://sites.dartmouth.edu/teachingresources/2025/08/01/learning-objectives-outcomes/
14. Dos, Don'ts, and Cautions in SLO Writing I. Limit each SLO to one measureable learning outcome (i.e., one action verb). For e, accessed March 30, 2026, https://www.ubalt.edu/_migrated-binaries/ubalt.edu/institutional_effectiveness/uploads/documents/assessment_handbook/Dos%20Donts%20and%20Cautions%20in%20SLO%20Writing.pdf
15. Learning Outcome vs. Learning Objective - UCLA Health, accessed March 30, 2026, https://www.uclahealth.org/sites/default/files/documents/ce-LearningOutcome-v-LearningObjective-052016.pdf
16. Outcomes, Objectives, and Alignment Demystified - UCATT - The University of Arizona, accessed March 30, 2026, https://ucatt.arizona.edu/news/outcomes-objectives-and-alignment-demystified
17. Writing and Assessing Learning Outcomes, accessed March 30, 2026, https://www.qqi.ie/sites/default/files/2024-02/prof-phil-newton-rethinking-assess

ment-of-learning-outcomes-in-he-slides.pdf
18. Writing Learning Outcomes, accessed March 30, 2026, https://www.qqi.ie/sites/default/files/2024-02/prof-phil-newton-rethinking-assessment-of-learning-outcomes-in-fet.pdf
19. Writing learning outcomes for GIHE courses - Library Glion, accessed March 30, 2026, https://library.glion.edu/celt/writing-learning-outcomes-for-gihe-courses/
20. Learning Outcomes | Centre for Teaching and Learning - University of Alberta, accessed March 30, 2026, https://www.ualberta.ca/en/centre-for-teaching-and-learning/teaching-support/preparation/learning-outcomes.html
21. Writing Student Learning Outcomes | Assessment | Center for Teaching Excellence, accessed March 30, 2026, https://miamioh.edu/centers-institutes/center-for-teaching-excellence/assessment/writing-student-learning-outcomes.html
22. Learning Objectives - Eberly Center - Carnegie Mellon University, accessed March 30, 2026, https://www.cmu.edu/teaching/designteach/design/learningobjectives.html
23. Teaching & Learning Resources - Eberly Center - Carnegie Mellon University, accessed March 30, 2026, https://www.cmu.edu/teaching/resources/index.html
24. Align Assessments, Objectives, Instructional Strategies - Eberly Center, accessed March 30, 2026, https://www.cmu.edu/teaching/assessment/basics/alignment.html
25. program-curriculum-mapping-instructions ... - BYU-Idaho, accessed March 30, 2026, https://brightspotcdn.byui.edu/33/70/6c61afd24cfe89ad8f3ce2d92cc5/program-curriculum-mapping-instructions-byui.docx
26. Guide to curriculum mapping: A systematic process for program ..., accessed March 30, 2026, https://educational-innovation.sydney.edu.au/teaching@sydney/curriculum-mapping-for-program-redesign/
27. Curriculum Mapping in Higher Education: Definition, Importance, Process, Software and Examples, accessed March 30, 2026, https://ems-works.com/blog/content/curriculum-mapping-in-higher-education-definition-importance-process-software-and-examples/
28. Ten Steps for Creating Academic Maps - Complete College America, accessed March 30, 2026, https://completecollege.org/news/ten-steps-for-creating-academic-maps/
29. Assessment and Standards, Revised 2022 - Quality and ..., accessed March 30, 2026, https://www.qqi.ie/sites/default/files/2022-09/assessment_and_standards-revised-2022.pdf
30. Direct Versus Indirect Assessment of Student Learning | Center for ..., accessed March 30, 2026, https://www.niu.edu/citl/resources/guides/direct-versus-indirect-assessment-of-student-learning.shtml

31. Direct and Indirect Assessment of Student Learning - Colorado State University, accessed March 30, 2026, https://assessment.colostate.edu/2025/01/07/direct-and-indirect-assessment-of-student-learning/
32. Direct Versus Indirect Assessment of Student Learning | Feedback & Grading | Teaching Guides - DePaul University Resources, accessed March 30, 2026, https://resources.depaul.edu/teaching-commons/teaching-guides/feedback-grading/Pages/direct-assessment.aspx
33. Choose a Method to Collect Data/Evidence | Assessment and Curriculum Support Center - University of Hawaii at Manoa, accessed March 30, 2026, https://manoa.hawaii.edu/assessment/resources/choose-a-method-to-collect-data-evidence/
34. Assessments - Eberly Center - Carnegie Mellon University, accessed March 30, 2026, https://www.cmu.edu/teaching/designteach/syllabus/newcourse/assessments.html
35. Program-level Assessment Tools - Eberly Center - Carnegie Mellon University, accessed March 30, 2026, https://www.cmu.edu/teaching/assessment/assessprogram/tools/
36. Rubrics | Resources | Instructional Design Studio - Seattle University, accessed March 30, 2026, https://www.seattleu.edu/instructional-design-studio/resources/rubrics/
37. VALUE Rubrics - AAC&U, accessed March 30, 2026, https://www.aacu.org/value/rubrics
38. VALUE Scoring Collaborative - AAC&U, accessed March 30, 2026, https://www.aacu.org/value/value-scoring-collaborative
39. VALUE Rubrics - Integrative Learning - AAC&U, accessed March 30, 2026, https://www.aacu.org/value/rubrics/value-rubrics-integrative-and-applied-learning
40. \[PDF\] Ireland - Cedefop, accessed March 30, 2026, https://www.cedefop.europa.eu/en/print/pdf/node/126101
41. a case study of implementing curriculum outcomes - Australian Association for Research in Education, accessed March 30, 2026, https://aare.edu.au/data/publications/1995/bradl95027.pdf
42. (PDF) (National Forum, 2017) Enhancing Programme Approaches to Assessment and Feedback in Irish Higher Education: Case Studies, Commentaries and Tools. - ResearchGate, accessed March 30, 2026, https://www.researchgate.net/publication/320955902_National_Forum_2017_Enhancing_Programme_Approaches_to_Assessment_and_Feedback_in_Irish_Higher_Education_Case_Studies_Commentaries_and_Tools
43. enhancing programme approaches to assessment and feedback in irish higher education: case studies, commentaries and tools, accessed March 30, 2026, https://www.teachingandlearning.ie/wp-content/uploads/Final-Programme-Assessment-Resource-with-doi-070717-1.pdf
44. Case Study Analysis as an Effective Teaching Strategy: Perceptions of

Undergraduate Nursing Students From a Middle Eastern Country - PMC, accessed March 30, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC8640316/
45. Common Problems with Learning Outcomes - WordPress.com, accessed March 30, 2026, https://afocusonlearningoutcomes.files.wordpress.com/2014/06/common-problems.pdf
46. Learning Outcomes Done Right: 9 Common Mistakes. Examples, solutions, and impact on your marketing. - Gerta.eu, accessed March 30, 2026, https://gerta.eu/learning-outcomes-done-right-9-common-mistakes-examples-solutions-and-impact-on-your-marketing/